

Linear Algebra – Homework 4

Sections 3.1–3.3 · Determinants, Cramer's Rule, and Adjugate Matrices · Full solutions

Problem 1

A, B are 3×3 with $\det A = -2$, $\det B = 3$. Compute each without finding entries.

- (a) $\det(AB) = \det A \cdot \det B = (-2)(3) = -6$
- (b) $\det(A^T) = \det A = -2$
- (c) $\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{-2} = -\frac{1}{2}$
- (d) $\det(4A) = 4^3 \det A = 64(-2) = -128$ (scalar c times an $n \times n$ matrix gives factor c^n ; here $n = 3$)
- (e) $\det(A^2 B^{-1}) = \det(A^2) \det(B^{-1}) = (\det A)^2 \cdot \frac{1}{\det B} = (-2)^2 \cdot \frac{1}{3} = \frac{4}{3}$

Problem 2

A is 3×3 , $\det A = 5$. B from A by: $R_1 \leftrightarrow R_3$, then $R_2 \leftarrow -2R_2$, then $R_3 \leftarrow R_3 + 7R_1$.

Track the determinant through each operation:

- $R_1 \leftrightarrow R_3$: row swap $\Rightarrow$ multiply by -1 . Det becomes -5 .
- $R_2 \leftarrow -2R_2$: multiplying one row by $-2 \Rightarrow$ multiply det by -2 . Det becomes $(-5)(-2) = 10$.
- $R_3 \leftarrow R_3 + 7R_1$: adding a multiple of one row to another leaves det unchanged. Det stays 10.

$\det B = 10$.

Problem 3

Operations on 3×3 A : $R_1 \leftrightarrow R_3$, $R_2 \leftarrow 3R_2$, $R_3 \leftarrow R_3 - 2R_1$, producing $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}$.

Find $\det A$ without reconstructing A .

Each operation multiplies the determinant by a known factor: swap $\times(-1)$, row-scale by 3×3 , row replacement $\times 1$. Hence

$$\det U = \det A \cdot (-1) \cdot 3 \cdot 1 = -3 \det A.$$

U is upper triangular, so $\det U = 2 \cdot 3 \cdot 2 = 12$. Therefore $12 = -3 \det A$.

$\det A = -4$.

Problem 4

True/False, with justification.

- (a) **False.** Determinant is not linear in matrix addition. Counterexample: $A = I_2$, $B = I_2$. Then $\det(A + B) = \det(2I_2) = 4$ but $\det A + \det B = 1 + 1 = 2$.

- (b) **False.** $\det A = 0$ means A is singular, but it need not have a zero row. Counterexample:
 $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\det A = 0$, no zero row.
- (c) **True.** $\det(AB) = \det A \det B = 0$, so at least one factor is 0.
- (d) **False.** Many distinct matrices share a determinant. Counterexample: $A = I_2$, $B = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$;
both have $\det = 1$ but $A \neq B$.
- (e) **True.** Adding a multiple of one row to another is a row replacement, which leaves the determinant unchanged.

Problem 5

$A_t = \begin{bmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}$. Use determinant properties and row operations (not a full cofactor expansion) to find $\det(A_t)$; for what value(s) of t is A_t invertible?

Apply row-replacement operations, which leave the determinant unchanged. Subtract the first row from the other two rows:

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - R_1 : \quad A_t \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & t-1 & 0 \\ 0 & 0 & t-1 \end{bmatrix}.$$

Each step is of the form $R_i \leftarrow R_i + kR_1$ (a row replacement), so $\det$ is unchanged. The result is upper triangular, and the determinant of a triangular matrix is the product of its diagonal entries:

$$\det(A_t) = 1 \cdot (t-1) \cdot (t-1) = (t-1)^2.$$

A_t is invertible iff $\det(A_t) \neq 0$, i.e. $t-1 \neq 0$.

$\det(A_t) = (t-1)^2$; A_t is invertible exactly when $t \neq 1$.

Why $t = 1$ fails: every row becomes $(1, 1, 1)$, so the rows are linearly dependent and $\det A_1 = 0$. This is exactly the coefficient matrix of Problem 22, where Cramer's Rule therefore breaks down at $t = 1$.

Problem 6

$\det \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 0 & 1 & 1 & 2 \end{bmatrix}$ by reducing to upper triangular form.

$R_2 \leftarrow R_2 - 2R_1$: $[0, 1, 1, 1]$; $R_3 \leftarrow R_3 - R_1$: $[0, 1, 2, 3]$.

Now $R_3 \leftarrow R_3 - R_2$: $[0, 0, 1, 2]$; $R_4 \leftarrow R_4 - R_2$: $[0, 0, 0, 1]$.

All operations are row replacements, so the determinant is unchanged. The resulting matrix is upper triangular:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$\det = 1 \cdot 1 \cdot 1 \cdot 1 = 1.$$

Problem 7

$$\det \begin{bmatrix} 3 & 0 & 2 \\ 0 & -1 & 4 \\ 5 & 0 & 1 \end{bmatrix} \text{ via cofactor expansion; state the chosen row/column.}$$

Choice: column 2, because it has two zeros: entries $(1, 2) = 0$, $(2, 2) = -1$, $(3, 2) = 0$.

$$\det = (-1)^{2+2}(-1) M_{22} = (-1) \cdot \det \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix} = (-1)(3 \cdot 1 - 2 \cdot 5) = (-1)(-7) = 7.$$

$$\det = 7.$$

Problem 8

$$\det \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 \\ 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 5 \end{bmatrix} \text{ via cofactor expansion.}$$

Expand along column 2 (only entry $(2, 2) = 3$ is nonzero; cofactor sign $(-1)^{2+2} = +1$):

$$\det = 3 \cdot M_{22}, \quad M_{22} = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}.$$

Expand M_{22} along its 3rd row (only $(3, 3) = 5$, sign $(-1)^{3+3} = +1$):

$$M_{22} = 5 \cdot \det \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} = 5(8 - 1) = 35.$$

$\det = 3 \cdot 35 = 105$. **The zeros let almost every cofactor term vanish.**

Problem 9

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \\ 2 & 1 & -1 \end{bmatrix}. \text{ Find } M_{12}, C_{12}, M_{23}, C_{23}, M_{31}, C_{31}, \text{ then } \det A \text{ by expanding the 2nd row.}$$

$$M_{12} = \det \begin{bmatrix} 0 & 5 \\ 2 & -1 \end{bmatrix} = 0(-1) - 5(2) = -10; \quad C_{12} = (-1)^{1+2} M_{12} = 10.$$

$$M_{23} = \det \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} = 1(1) - (-2)(2) = 5; \quad C_{23} = (-1)^{2+3} M_{23} = -5.$$

$$M_{31} = \det \begin{bmatrix} -2 & 3 \\ 4 & 5 \end{bmatrix} = (-2)(5) - 3(4) = -22; C_{31} = (-1)^{3+1}M_{31} = -22.$$

Second-row expansion needs C_{21}, C_{22} too:

$$M_{21} = \det \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix} = 2 - 3 = -1 \Rightarrow C_{21} = 1;$$

$$M_{22} = \det \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = -1 - 6 = -7 \Rightarrow C_{22} = -7.$$

$$\det A = 0 \cdot C_{21} + 4 \cdot C_{22} + 5 \cdot C_{23} = 0 + 4(-7) + 5(-5) = -28 - 25 = -53.$$

$$M_{12} = -10, C_{12} = 10, M_{23} = 5, C_{23} = -5, M_{31} = -22, C_{31} = -22; \det A = -53.$$

Problem 10

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 2 \\ 0 & 4 & 5 \end{bmatrix}.$$

(a) Six-term (diagonal) formula:

$$\det A = 2[(-1)(5) - 2(4)] - 0 + 1[3(4) - (-1)(0)] = 2(-13) + 12 = -26 + 12 = -14.$$

(b) Cofactor expansion along column 2 (entries 0, -1, 4):

$$M_{22} = \det \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix} = 10 \Rightarrow C_{22} = 10; M_{32} = \det \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = 1 \Rightarrow C_{32} = -1.$$

$$\det A = 0 \cdot C_{12} + (-1) \cdot 10 + 4 \cdot (-1) = -10 - 4 = -14.$$

(c) Both compute the same quantity: the determinant is a single well-defined number for the matrix, given by the Leibniz/multilinear-alternating definition. The “six-term” formula *is* the cofactor expansion along the first row fully written out; cofactor expansion is just a recursive application of that same definition. Different paths must yield the same value.

$\det A = -14$ **both ways**.

Problem 11

General determinant formula: each term has exactly one entry from each row and each column.

- (a) A 4×4 determinant has one term per permutation of $\{1, 2, 3, 4\}$: **4!** = **24** terms.
- (b) $a_{11}a_{22}a_{33}a_{44}$ corresponds to column order $(1, 2, 3, 4)$, a valid permutation (one from each row and column). $a_{11}a_{22}a_{34}a_{44}$ uses column 4 twice and omits column 3 — not a permutation — so it cannot appear.
- (c) Product $a_{12}a_{24}a_{31}a_{43}$ has column order $(2, 4, 1, 3)$. Count inversions in $(2, 4, 1, 3)$: $(2, 1), (4, 1), (4, 3) \rightarrow 3$ inversions, an odd number, so the term carries a **minus** sign.

Problem 12

Rows of A are r_1, r_2, r_3 , $\det A = 6$. B has rows $r_1 + 2r_2$, $-3r_2$, $r_3 - 5r_2$. Find $\det B$ without entries.

Using multilinearity of the determinant in the rows,

$$\det B = \det(r_1 + 2r_2, -3r_2, r_3 - 5r_2) = \det(r_1, -3r_2, r_3 - 5r_2) + 2 \det(r_2, -3r_2, r_3 - 5r_2).$$

First term: linearity in row 3 gives $\det(r_1, -3r_2, r_3) + (-5) \det(r_1, -3r_2, r_2)$. The second piece has rows 2 and 3 both proportional to $r_2 \rightarrow 0$. So $= (-3) \det(r_1, r_2, r_3) = -3(6) = -18$.

Second term: both pieces contain rows 1 and 2 proportional to $r_2 \rightarrow 0$.

$$\det B = -18.$$

Equivalently, the row transformation matrix is $\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 0 \\ 0 & -5 & 1 \end{bmatrix}$, whose determinant is -3 , so

$$\det B = (-3) \det A = -18.$$

Problem 13

Explain why $a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$, but $a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} = \det A$.

Let B be A with its second row replaced by the first row of A . Then B has two identical rows (row 1 = row 2), so $\det B = 0$. Expanding $\det B$ along its second row:

$$\det B = b_{21}C'_{21} + b_{22}C'_{22} + b_{23}C'_{23} = a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23}$$

because $b_{2j} = a_{1j}$ and deleting the (replaced) second row leaves rows 1 and 3 of A , so the minors — hence cofactors C'_{2j} — equal A 's C_{2j} . Thus the sum is 0.

For the second identity, $a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23}$ is exactly the cofactor expansion of $\det A$ along its actual second row, so it equals $\det A$ by definition.

Problem 14

$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. Find the cofactor matrix C , $\text{adj}(A)$, and $\det A$.

$$\det A = 1(3 \cdot 1 - 1 \cdot 0) - 2((-1) \cdot 1 - 1 \cdot 2) + 0 = 3 - 2(-3) = 9.$$

Cofactors:

$$C_{11} = 3, C_{12} = 3, C_{13} = -6; \quad C_{21} = -2, C_{22} = 1, C_{23} = 4; \quad C_{31} = 2, C_{32} = -1, C_{33} = 5.$$

$$C = \begin{bmatrix} 3 & 3 & -6 \\ -2 & 1 & 4 \\ 2 & -1 & 5 \end{bmatrix}, \quad \text{adj}(A) = C^T = \begin{bmatrix} 3 & -2 & 2 \\ 3 & 1 & -1 \\ -6 & 4 & 5 \end{bmatrix}.$$

$\det A = 9$ (and $A \text{adj}(A) = 9I$, confirming the cofactors).

Problem 15

Using Problem 14, $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$; verify one row-by-column product in AA^{-1} .

$$A^{-1} = \frac{1}{9} \begin{bmatrix} 3 & -2 & 2 \\ 3 & 1 & -1 \\ -6 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 1/3 & -2/9 & 2/9 \\ 1/3 & 1/9 & -1/9 \\ -2/3 & 4/9 & 5/9 \end{bmatrix}.$$

Verify entry (2, 2) of AA^{-1} : row 2 of A is $(-1, 3, 1)$, column 2 of A^{-1} is $(-2/9, 1/9, 4/9)$:

$$(-1)(-2/9) + 3(1/9) + 1(4/9) = 2/9 + 3/9 + 4/9 = 9/9 = 1.$$

Confirmed (off-diagonal entries similarly give 0).

Problem 16

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Find $\operatorname{adj}(A)$ and verify $A \operatorname{adj}(A) = \det(A)I_2$; explain why $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$ needs $\det A \neq 0$.

$\operatorname{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Then

$$A \operatorname{adj}(A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)I_2 = \det(A)I_2.$$

The formula $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$ requires division by $\det A$, which is only valid when $\det A \neq 0$ (and that condition is exactly what makes A invertible).

Problem 17

A student: “If $\det A = 0$ then $A \operatorname{adj}(A) = 0$, therefore $\operatorname{adj}(A) = 0$.” Explain the error; give a 2×2 example with $\det A = 0$ but $\operatorname{adj}(A) \neq 0$.

From $A \operatorname{adj}(A) = \det(A)I = 0$ we may only conclude that each *column* of $\operatorname{adj}(A)$ lies in the null space of A . A nonzero matrix times a nonzero matrix can be the zero matrix; $\operatorname{adj}(A) = 0$ does not follow.

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\det A = 0$. Its adjugate is $\operatorname{adj}(A) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \neq 0$. Indeed $A \operatorname{adj}(A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Problem 18

$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $\operatorname{adj}(A) = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$. (a) Find $\det A$ via $A \operatorname{adj}(A) = \det(A)I$. (b)

Find A^{-1} .

(a) $A \operatorname{adj}(A) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2I$. Hence $\det A = 2$.

$$(b) \ A^{-1} = \frac{1}{2} \text{adj}(A) = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \end{bmatrix}.$$

Problem 19

$$\text{Cramer's Rule for } \begin{cases} 2x - y = 1 \\ 3x + 2y = 8 \end{cases}.$$

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}, \ b = \begin{bmatrix} 1 \\ 8 \end{bmatrix}, \ \det A = 2(2) - (-1)(3) = 7.$$

$$A_1 = \begin{bmatrix} 1 & -1 \\ 8 & 2 \end{bmatrix}, \ \det A_1 = 1(2) - (-1)(8) = 10 \Rightarrow x = 10/7.$$

$$A_2 = \begin{bmatrix} 2 & 1 \\ 3 & 8 \end{bmatrix}, \ \det A_2 = 2(8) - 1(3) = 13 \Rightarrow y = 13/7.$$

$$(x, y) = (10/7, 13/7).$$

Problem 20

$$\begin{cases} x + 2y - z = 6 \\ 2x + y + z = 3 \\ x - y + z = -2 \end{cases}.$$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}, \ b = \begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix}.$$

$$\det A = 1(1 + 1) - 2(2 - 1) + (-1)(-2 - 1) = 2 - 2 + 3 = 3.$$

$$A_1 = \begin{bmatrix} 6 & 2 & -1 \\ 3 & 1 & 1 \\ -2 & -1 & 1 \end{bmatrix}: \det A_1 = 6(2) - 2(1) + (-1)(-1) = 12 - 10 + 1 = 3 \Rightarrow x = 1.$$

$$A_2 = \begin{bmatrix} 1 & 6 & -1 \\ 2 & 3 & 1 \\ 1 & -2 & 1 \end{bmatrix}: \det A_2 = 1(5) - 6(1) + (-1)(-7) = 5 - 6 + 7 = 6 \Rightarrow y = 2.$$

$$A_3 = \begin{bmatrix} 1 & 2 & 6 \\ 2 & 1 & 3 \\ 1 & -1 & -2 \end{bmatrix}: \det A_3 = 1(-1) - 2(-7) + 6(-3) = 1 + 14 - 18 = -3 \Rightarrow z = -1.$$

$$(x, y, z) = (1, 2, -1).$$

Problem 21

$$\begin{cases} 2x_1 - x_2 + x_3 = 5 \\ x_1 + 2x_2 = -1 \\ 3x_1 + x_2 + 2x_3 = 6 \end{cases}. \text{ Use Cramer's Rule to find only } x_2.$$

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix}. \det A = 2(4) - (-1)(2) + 1(1 - 6) = 8 + 2 - 5 = 5.$$

$$A_2 = \begin{bmatrix} 2 & 5 & 1 \\ 1 & -1 & 0 \\ 3 & 6 & 2 \end{bmatrix} \text{ (replace column 2 with } b\text{).}$$

$$\det A_2 = 2(-2) - 5(2) + 1(6 + 3) = -4 - 10 + 9 = -5.$$

$$x_2 = \det A_2 / \det A = -5/5 = -1.$$

Problem 22

$$\begin{cases} x + y + z = 1 \\ x + ty + z = 1 \\ x + y + tz = 1 \end{cases}.$$

- (a) Coefficient matrix $M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}$. Using the same subtract-the-first-row idea as in Problem 5:

$R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$ gives upper triangular $\text{diag}(1, t-1, t-1)$, so $\det M = (t-1)^2$. Cramer's Rule applies when $\det M \neq 0$, i.e. $t \neq 1$.

- (b) For $t \neq 1$: subtracting the first equation from the 2nd and 3rd gives $(t-1)y = 0$ and $(t-1)z = 0$, hence $y = z = 0$, and then $x = 1$. So $(x, y, z) = (1, 0, 0)$.
- (c) When $t = 1$ all three equations become $x + y + z = 1$; the system collapses to one equation, so it has **infinitely many solutions** (a plane). Cramer's Rule fails because $\det M = 0$.

Problem 23

Show $x_j = \frac{\det(A_j)}{\det A}$ by connecting the matrix product $\text{adj}(A)b$ with a cofactor expansion.

$$x = A^{-1}b = \frac{1}{\det A} \text{adj}(A)b. \text{ The } j\text{-th component is } x_j = \frac{1}{\det A} (\text{adj}(A)b)_j.$$

Row j of $\text{adj}(A)$ equals column j of the cofactor matrix C , namely $(C_{1j}, C_{2j}, \dots, C_{nj})$. So

$$(\text{adj}(A)b)_j = \sum_{i=1}^n C_{ij} b_i.$$

Now A_j is A with column j replaced by b . Expanding $\det(A_j)$ along column j :

$$\det(A_j) = \sum_{i=1}^n (A_j)_{ij} (C_j)_{ij} = \sum_{i=1}^n b_i C_{ij},$$

because the cofactors $(C_j)_{ij}$ are computed by deleting row i and column j — column j is the one we replaced, so they are unchanged from A 's cofactors C_{ij} . Thus $(\text{adj}(A)b)_j = \det(A_j)$, and

$$x_j = \frac{\det(A_j)}{\det A}.$$

Problem 24

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. A student found $C = \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$ but wrote $\text{adj}(A) = C$.

(a) The error: $\text{adj}(A)$ is the **transpose** of the cofactor matrix, not the cofactor matrix itself. (The student's C is correct: $C_{11} = 4$, $C_{12} = -3$, $C_{21} = -2$, $C_{22} = 1$.)

(b) Correct adjugate: $\text{adj}(A) = C^T = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$.

(c) $AC = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 4 & -5 \end{bmatrix}$ — not of the form $\det(A)I$.

$$A \text{adj}(A) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \det(A)I.$$

$A \text{adj}(A)$ **has the form** $\det(A)I$ (**here** $\det A = -2$).

Problem 25

A is a 3×3 real matrix with $A^T = -A$ (skew-symmetric). Prove $\det A = 0$ without expanding.

$$\det(A) = \det(A^T) = \det(-A) = (-1)^3 \det(A) = -\det(A).$$

So $\det(A) = -\det(A)$, hence $2\det(A) = 0$, and therefore $\det(A) = 0$.

Key facts used: $\det(A^T) = \det(A)$ and $\det(cA) = c^n \det(A)$ with $n = 3$. (For odd dimension a real skew-symmetric matrix is always singular; in even dimensions it need not be.)